Response to Reviewers

Manuscript: *Adapter Scaling Trade-off and Timestep-Conditioned Scheduling in Multi-view Diffusion Texture Generation* (recommended by CAD/Graphics 2026 for publication in *Computers & Graphics*)

Dear Editors and Reviewers,

We thank the CAD/Graphics 2026 program committee for recommending this manuscript, upon revision, for publication in *Computers & Graphics* (VSI: CAG_SS_CAD/Graphics 2026), and we thank all three reviewers for their careful and constructive comments. All main TCAS results and conclusions are retained in the revised manuscript; following the reviewers' suggestions, we have added new analyses, documentation, and controlled experiments, as detailed below. For transparency, we note that the learned FAC extension, initially reported positively under our earlier protocol, is now re-examined under a strictly paired, strictly disjoint protocol with a negative outcome (Section 4.6; see our response to Comment 1 of Reviewer 1). In the marked manuscript, the corresponding changes are highlighted in blue.

Best regards

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Response to Reviewer 1

1. The data splits and the FAC evaluation are unclear; FAC requires a fully described training objective and separate training, tuning, and test sets.

Response: Thank you for this helpful comment. We revised the manuscript in two aspects.

(1) We clarified the data splits and disclosed the probe/holdout structure. Section 4.1 now documents the complete data pipeline: all objects are drawn from the Objaverse 3D asset collection; ground-truth multi-view images are rendered offline with Blender (Cycles engine, 17 views at 512×512, white background); the adapter training pool contains 1,118 rendered objects and the evaluation pool contains 300 rendered objects, and we verified that the two identifier sets intersect in zero objects, so no evaluation object was seen during adapter training. We additionally disclose that the 24-object probe set (obj_0000–obj_0023) is part of the 300-object pool, and every transfer claim

is now also verified on the strictly disjoint 276-object holdout (obj_0024–obj_0299). The 50-object scale-sweep set, the 26-object texture-audit set, the 50-object CLIP-IQA subset, and the 30-object preference-study subset are all subsets of the 300-object pool, and no training object appears in any of them. (Please see the "Implementation and dataset details" paragraph in Section 4.1 and Section 4.7 of the revised manuscript.)

(2) We fully described the FAC training configuration and added a controlled re-examination. Section 4.6 now specifies the complete configuration: roughly 6×10^3 trainable parameters (the LTAG/GSG/FSC controllers only; the base model, VAE, and adapter remain frozen), the same enhanced denoising objective as the base adapter (a latent noise-prediction MSE with 1.7× foreground and 1.3× edge spatial reweighting plus a latent-SSIM term), AdamW with peak learning rate 10^-4 (100 warm-up steps) for 2,000 training steps, and a training pool strictly disjoint from all evaluation objects (no evaluation object is used for training or model selection). We further report controls spanning the most favourable training conditions we could construct: a warm-start control whose gate is initialized at the exact C3 schedule and left untrained is statistically indistinguishable from TCAS (paired ΔFG-PSNR = −0.18 dB on the control subset, 95% CI crossing zero, validating the evaluation path), whereas every trained configuration degrades fidelity, with paired deficits ranging from −1.1 to −2.9 dB FG-PSNR (win rates at most 16%) and increasing monotonically with the amount of controller training. Preserving the temporal envelope is therefore necessary but not sufficient: the limitation lies in the learned-modulation training signal itself, which trades foreground fidelity for denoising-objective loss. The training-free TCAS remains the main and sole method of the paper. For transparency, we note that an earlier version of this extension, evaluated on the main adapter under our initial protocol before the strictly paired and strictly disjoint training controls were adopted, had suggested a positive FAC gain; that preliminary comparison differed in both evaluation protocol and adapter instance, and under the strictly paired, strictly disjoint protocol no trained configuration replicates the gain, so the earlier comparison is superseded by the controlled re-examination. (Please see Section 4.6 of the revised manuscript; the complete dose–response table over all seven trained configurations is provided in the supplementary material.)

2. Provide the exact implementation details and checkpoints, conditioning, resolution, sampler, scheduler, prompts or references.

Response: Thank you for this suggestion. We added a dedicated "Implementation and dataset details" subsection (Section 4.1.3) with a consolidated reproducibility reference. It specifies: the base pipeline (an MVPainter-style geometry-controlled multi-view diffusion model built on a joint six-view latent UNet, used frozen together with its VAE); the adapter input (a five-channel geometric input of a 3-channel normal map, a 1-channel depth map, and a 1-channel foreground mask) and its multi-scale injection into multiple UNet resolution levels; the adapter checkpoint used for all tables and its scale semantics (the requested scale is applied multiplicatively at every injected layer); the sampler and scheduler (Euler discrete, 50 denoising steps); the fixed random seed (42) shared by all schedules within an experiment, so all variants share the same initial latents; the six target views and the ground-truth rendering configuration; and the exact metric implementations (foreground-masked FG-SSIM/Edge-SSIM/PSNR, edge masks derived from depth/normal discontinuities, LPIPS with the AlexNet backbone, and foreground Laplacian-variance/RGB-std/gradient-magnitude diagnostics computed under identical views and masks). The pipeline is reference-image-conditioned; no text prompts are used. In addition, the supplementary material now contains a consolidated reproducibility table listing the data source and subsets, the exact base-pipeline and adapter checkpoint identifiers, the scheduler, steps, seed, conditioning, resolution, and metric implementations. (Please see Section 4.1 of the revised manuscript and the reproducibility table in the supplementary material.)

3. Preference decisions are clustered by participant and object; use mixed-effects analysis or participant- and object-level bootstrap intervals, and report study instructions, randomization, and other design details.

Response: Thank you for this valuable comment, which we fully addressed in two respects.

(1) We now report the complete study design in Section 4.5: a blinded three-alternative forced-choice (3AFC) preference study on 30 objects drawn from the evaluation pool with 24 participants; for each object and each of the three criteria (texture naturalness, shape consistency, overall quality), participants viewed the anonymized outputs of the three conditions side by side; the left-to-right condition order was randomized independently per trial; participants were not informed which method produced which image or how many conditions existed; instructions asked participants to select the best image for the stated criterion only; and every participant judged all 30 objects under all three criteria, yielding 720 valid first-choice decisions per criterion.

(2) We adopted cluster-aware inference. Because first-choice decisions are clustered within both participants and objects, we report 95% confidence intervals obtained by a two-level cluster bootstrap (resampling participants and objects with replacement, 10,000

resamples). C3's overall-quality share is 58.1% (CI [50.1%, 65.8%]), exceeding the conservative and aggressive baselines by +33.6 points (CI [19.3, 47.2]) and +40.6 points (CI [29.4, 51.8]); both differences exclude zero, so the overall-preference advantage of C3 is significant. By contrast, the baselines' single-criterion advantages are not significant (texture naturalness: $s = 1.25$ over C3 +6.7 points, CI [−7.9, 21.5]; shape consistency: $s = 2.50$ over C3 +5.3 points, CI [−6.7, 17.9]). Overall quality is therefore the only criterion with a significant preference, supporting the interpretation of TCAS as the perceptually preferred balance rather than a per-criterion maximum. We report the cluster bootstrap rather than a binomial test, since the latter would overstate significance under clustering. (Please see Section 4.5 of the revised manuscript.)

Response to Reviewer 2

1. Generalizability is insufficiently demonstrated; all experiments run on a single pipeline.

Response: Thank you for this important comment. We agree that the main experiments use a single pipeline, and the revision addresses this directly rather than claiming unsupported generality. New Section 4.7 ("Adapter-Dependence of High-Scale Failure Modes") reports controlled mechanism probes on additional adapter checkpoints of the same architecture under an identical inference protocol: across three regimes spanning a spectrum of trained residual magnitudes, uniform high scale ($s = 2.50$) produced (i) texture flattening with the main adapter studied in the paper, (ii) high-frequency artifact amplification with a strong-residual adapter under per-layer caps, and (iii) no appreciable effect with a deliberately weakened adapter. A per-layer scale-cap ablation reproduces both directions on a single checkpoint, isolating the inference-time scale semantics as the determining factor for the failure direction; a residual-magnitude sweep (scaling the adapter's output projection from 0.1× to 3×) shows that the intervention strength is monotonic in the trained residual magnitude; and a cross-adapter stage ablation shows that even the most sensitive denoising stage is adapter-dependent. What remains stable is stated precisely: in every regime where scaling had a non-negligible effect, the low–high–low temporal schedule (C3) yielded the best shape–texture balance, because concentrating correction in the middle stage exploits the stage-dependent division of labor observed in our experiments (global layout → geometry refinement → texture synthesis), which we verify across the tested adapter regimes rather than claim as a model-independent property. We explicitly acknowledge in Section 5 (Limitations) that the evidence remains within one architecture family, that the schedule has not yet been validated on a different geometry-conditioned multi-view diffusion backbone, and that cross-backbone validation is future work; the contribution is now framed as a conditional structural rule with re-estimable stage utilities rather than a universal optimum. (Please see Sections 4.7 and 5 of the revised manuscript.)

2. TCAS is heuristic; the boundaries and values are not explained.

Response: Thank you for pointing this out. The revision replaces the heuristic framing with a formal derivation plus robustness checks. First, Section 3.3 introduces a measurable Correction–Alignment–Interference (CAI) stage-utility model, stated as Proposition 1 with a proof: a two-stage, guardrailed epsilon-optimal selection rule — first maximize the fidelity objective over the admissible schedules, then minimize the texture-deviation risk among schedules whose fidelity is within epsilon = 0.1 dB of the maximum. The measured utilities select exactly the low–high–low assignment (U_e = −0.788 dB, CI [−1.420, −0.156], guardrail-violating; U_m = +0.574 dB, CI [0.391, 0.756] — above epsilon on its own, so the high scale must be retained in the middle stage; U_l = +0.077 dB, CI [0.006, 0.149] — CI lower bound +0.006 dB, far below epsilon, so the late stage is fidelity-equivalent under either assignment and the risk-dominant conservative scale is selected; these utilities are measured on the strong-residual, per-layer-capped adapter instance of Section 4.7, where per-stage effects are most visible, while the main adapter's 17-variant probe independently selects the same low–high–low assignment). The proposition is explicitly conditional on the measured adapter regime. Second, a boundary/peak sensitivity sweep (three high-scale windows × five peak scales, fifteen candidates on the 24-object probe protocol of the same strong-residual instance, Section 4.7) shows that the canonical middle-third window lies in a stable high-performance region (its 2.50 setting achieves 16.71 dB PSNR; the best setting in the same window, 2.75, achieves 16.79 dB but with worse texture diagnostics and lower FG-SSIM), so C3 is a robust middle-stage choice rather than a boundary-sensitive point estimate. Third, the probe variants already include timestep-conditioned alternatives that place the high scale in early, late, or broader windows (Table 2 and Fig. 7 of the revised manuscript; Table 2 shows eight representative variants), and the disjoint-holdout transfers described in our response to Reviewer 3 rule out the remaining generic schedules (linear warm-up, cosine bump, trapezoid, and Gaussian-peak). (Please see Sections 3.3 and 4.7 of the revised manuscript.)

3. The subset sources/construction/overlap and the FAC training details are insufficient.

Response: Thanks for pointing this out. The subset construction and the now-disclosed probe/holdout overlap are addressed in Section 4.1 (please see our response to Comment 1 of Reviewer 1, item (1)). The complete FAC training configuration and the controlled re-examination, including the warm-start control and its scoped interpretation, are reported in Section 4.6 (please see our response to Comment 1 of Reviewer 1, item (2)). TCAS remains the main and sole method of the paper.

Response to Reviewer 3

1. Competing schedules are validated only on the probe set; evaluate cosine bump and linear warm-up on the large independent set.

Response: Done, thank you for this suggestion. Linear warm-up and the cosine bump were regenerated at the full 300-object scale on the strong-residual, per-layer-capped adapter instance described in Section 4.7, and the transfer statistics are reported on the strictly disjoint 276-object holdout (obj_0024–obj_0299; Section 4.7): C3 improves PSNR over linear warm-up by +0.323 dB (95% CI [0.263, 0.386], 214/276 wins) while matching its structure (FG-SSIM difference −0.001, statistically tied), and over the cosine bump by +0.902 dB (95% CI [0.824, 0.981], 253/276 wins) with a higher FG-SSIM (+0.046). Together with the transferred trapezoid (+0.258 dB, CI [0.207, 0.308], 206/276 wins) and Gaussian-peak (+0.324 dB, CI [0.281, 0.367], 232/276 wins) schedules, the holdout ordering matches the probe: warm-up approaches C3 in structure but loses in signal fidelity, and the smooth bump loses in both because its effective high-scale duration is shorter than the full middle third. These statistics are now also summarized in a dedicated scheduling-transfer table (Table 9 in Section 4.7). (Please see Section 4.7 of the revised manuscript.)

2. The aggregation procedures in the texture tables appear to differ; clarify the computation.

Response: The reviewer is correct, and we thank them for the careful reading. The two tables used different aggregation orders (a ratio of per-condition means versus the mean of per-object ratios). The revised captions now state the rule explicitly: absolute foreground statistics are means over objects and views, and each ratio is the mean over objects of the per-object ratio relative to the corresponding $s = 1.25$ statistic (equal object weighting). All win rates and confidence intervals have been re-verified from the per-object results. (Please see the captions of the texture tables in Section 4.5 of the revised manuscript.)

3. Clarify the statistical reporting (±, paired test, bootstrap procedure).

Response: Thank you for this suggestion. Statistical reporting is now clarified at first use in Section 4.1 and in the relevant captions. Specifically, for CLIP-IQA: ± denotes the standard deviation across the 50 objects; the paired comparison is a two-sided paired t-test ($t(49) = 10.52$, $p \approx 3.6\times10^{-14}$ for C3 vs. $s = 2.50$); and the 95% confidence interval of the paired mean difference is constructed by object-level percentile bootstrap with 10,000 resamples ([0.0037, 0.0054], excluding zero). For the preference study, two-level (participant- and object-level) cluster-bootstrap intervals are reported in Section 4.5 (please see our response to Comment 3 of Reviewer 1). All other paired statements (stage utilities, holdout deltas) use object-level paired statistics with percentile bootstrap 95% CIs, and the procedure is stated once in Section 4.1.3. (Please see Sections 4.1 and 4.5 of the revised manuscript.)

4. Provide more complete dataset/implementation details; FAC training data, loss, optimization, parameter count, and the exclusion of the 300 validation objects.

Response: Thank you for this comment. The dataset and implementation details, including the verified zero train–eval overlap, are provided in Section 4.1 (please see our responses to Comments 1 and 2 of Reviewer 1). For FAC, Section 4.6 now specifies exactly what is requested: training data (the 1,118-object adapter training pool, strictly disjoint from all evaluation objects; no evaluation object is used for training or model selection), loss (the same enhanced denoising objective as the base adapter, specified in Section 4.6), optimization (AdamW, peak learning rate 10^-4, 2,000 steps), parameter count (roughly 6×10^3 trainable parameters), and the strictly paired evaluation with a warm-start control. The controlled outcome and its scoped interpretation are reported in the manuscript, and the training-free TCAS remains the main and sole method. (Please see Section 4.6 of the revised manuscript.)

We believe the revision is substantially strengthened, and we thank the reviewers again: their comments directly led to the CAI stage-utility model, the disjoint holdout protocol, the adapter-dependence analysis, and the cluster-aware statistical reporting.

Sincerely, The Authors